

High-Altitude Ground Stations: Challenges and Opportunities in All-Optical Satellite Networks

Benoit Coeugnet, Pablo G. Madoery, Juan A. Fraire, Jorge M. Finochietto, Halim Yanikomeroglu

Abstract—Low Earth orbit (LEO) satellite constellations are emerging as critical components of next-generation networks. They work with terrestrial systems to deliver high-throughput, low-latency connectivity globally. Direct-to-device communication and bulk data transport services are among the most promising use cases driving these constellations' adoption and economic potential. However, both use cases face significant technical and operational challenges. This paper introduces an architecture where high-altitude ground stations (HAGS) unlock the full potential of direct-to-device communications and all-optical LEO constellations. By enhancing the cost-efficiency of direct-to-device access and bulk data transport, HAGS enable the exclusive use of optical transceivers in LEO satellites, offering significant benefits in size, weight, power efficiency, cost, and data throughput. The paper further explores these opportunities through simulations that quantify the benefits of the proposed architecture.

Index Terms—Satellite Networks, LEO Constellations, Direct-to-Device, HAPS, HAGS, FSO.

I. INTRODUCTION

While the standardization process of 6G is still underway, it is becoming clear how it will distinguish itself from previous generations. Beyond aiming for a tenfold improvement in data rates and latency, 6G sets forth a transformative vision to achieve inclusive connectivity, ubiquitous coverage, and enhanced sustainability. Cost-effectively reaching these ambitious objectives requires a crucial focus on the seamless integration of terrestrial and non-terrestrial networks (NTN) [1]. Over the last few years, this focus and advancements in launch and satellite technologies have renewed interest in deploying prominent low-Earth orbit (LEO) satellite constellations.

One of the most critical drivers for deploying LEO constellations is providing broadband services to remote and underserved regions where terrestrial infrastructure is either unavailable or prohibitively expensive [2]. Among the major players in this domain, Starlink and OneWeb stand out. Starlink has launched approximately 7,000 satellites, representing roughly 60% of the total operational satellites in orbit as of 2024. OneWeb, with 618 satellites currently in orbit, also plays a significant role in expanding global broadband coverage. Meanwhile, Telesat Lightspeed and Kuiper are preparing to

enter the market with planned constellations of 198 and 3,236 satellites. This further underscores the growing momentum and demand for connectivity through LEO constellations.

In addition to broadband services, two key use cases for LEO constellations are *direct-to-device* communication and *bulk data transport*. Direct-to-device communication enables seamless connectivity for standard smartphones without additional hardware, extending network coverage to underserved areas. In-orbit bulk data transport focuses on efficiently handling and downloading large volumes of aggregated data from applications such as the Internet of Things (IoT) and Earth Observation (EO). This data is managed in space using storage, inter-satellite links (ISL) for opportunistic forwarding, and advanced routing strategies to optimize transmission to ground stations.

Both use cases face significant challenges with traditional architectures. A key obstacle to direct-to-device communication is overcoming the limitations of dynamic and long-range access channels to unmodified handheld devices, which currently exhibit low performance and scalability. For bulk data transport, the main challenge lies in the bottleneck created by the mismatch between the vast volumes of data aggregated by LEO satellites and the limited capacity for transferring that data to the ground through the feeder links.

To address these challenges, we propose an architecture that integrates high-altitude ground stations (HAGS) with all-optical LEO constellations. HAGS, a specialized subset of high-altitude platform stations (HAPS) [3], can shorten the access link distance and establish localized, shorter-range and static (Doppler-free) channels, thereby significantly enhancing direct-to-device scalability and performance. For bulk data transport, HAGS can effectively receive, store, and transmit data from passing-by LEO satellites through free-space optical (FSO) links while mitigating weather-related impairments.

As a result, our solution extends the performance and scalability of emerging direct-to-device services while enabling the deployment of all-optical LEO satellites, which rely exclusively on FSO transceivers. This approach reduces the size, weight, power (SWaP), and cost of the LEO satellite platform—especially when considering the higher kilo-cost of the satellite—compared to state-of-the-art configurations that require both radio-frequency (RF) and FSO transceivers onboard.

The remainder of the paper is structured as follows. Section II discusses the challenges posed by legacy satellite constellations. Section III presents the opportunities of HAGS in all-optical constellations. Section IV presents an evaluation of the proposed architecture, demonstrating and quantifying

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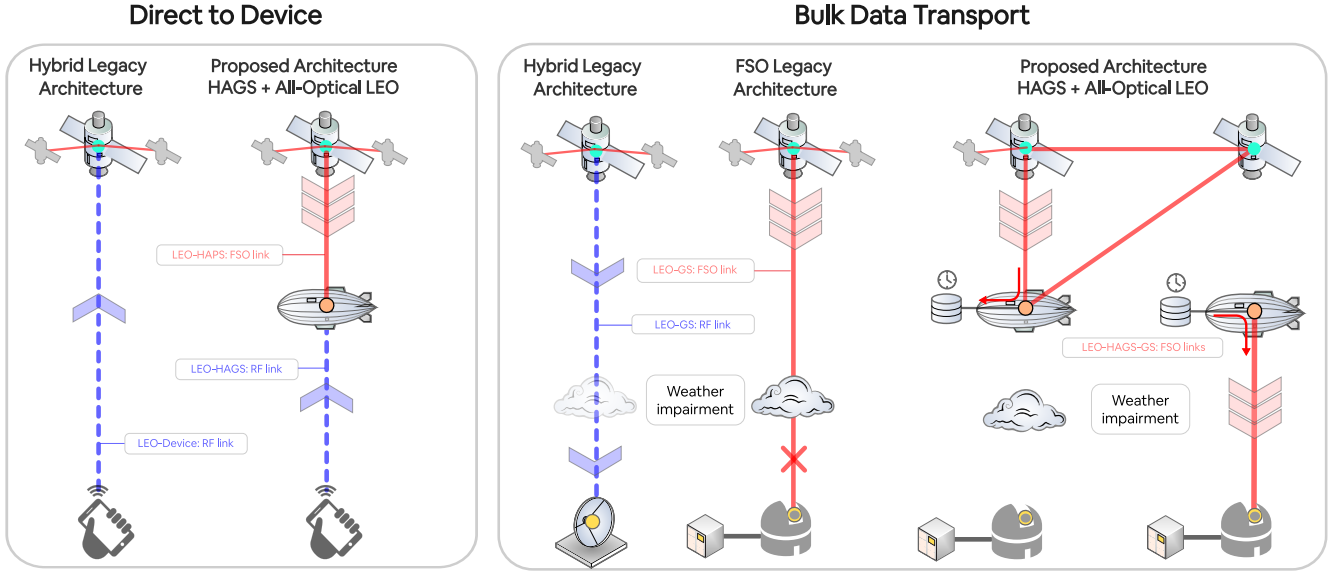


Fig. 1: Legacy and proposed architectures for LEO constellation use cases.

its advantages. Finally, Section V offers concluding remarks.

II. CHALLENGES IN LEO CONSTELLATION USE CASES

A. Direct-to-Device

Extending LEO constellation services to suburban and urban areas is an attractive option for broadening service adoption and increasing profitability. One key requirement for achieving widespread adoption is replacing proprietary user terminals (UTs) with standard, unmodified smartphones that do not require additional hardware. Fig. 1 illustrates this use case, where a smartphone directly connects to a LEO satellite, which can either act as a cellular base station in the sky (regenerative mode) or relay the received signals back and forth to a terrestrial cellular base station (transparent mode) [4].

Actors and Experiences: Several experiences prove the feasibility of direct-to-device. Lynk Global began providing intermittent text messaging services in 2022. The same year, Apple announced its partnership with Globalstar to provide emergency SOS services via satellite for the iPhone 14. AST SpaceMobile successfully conducted the first-ever two-way voice calls in 2023, and as of 2024, SpaceX has already deployed more than 180 Starlink satellites with direct-to-device capability.

Core Challenges: Despite its growing adoption and increasing popularity, several challenges are associated with this type of communication. Since LEO satellites orbit at altitudes between 300 and 2000 km, the free space path loss considerably decreases the power received by the satellite. Signals also experience significant Doppler shifts due to the high speed of LEO satellites (approximately 27,000 km/h) [5], and the limited availability of spectrum increasingly raises the probability of interference [6]. Additionally, since multiple users must share the communication channel, achievable throughputs are currently constrained to the Kbps range. Finally, tracking, handing over, and communicating with satellites hundreds of

kilometers away leads to higher battery consumption in mobile devices, further reducing the appeal of this approach.

B. Bulk Data Transport

Another prominent use case, illustrated in Fig. 1, uses LEO constellations for bulk data transport. This involves transporting large volumes of data from various devices, services, and applications to a ground-based infrastructure, which includes ground stations, gateways, and the broader Internet.

Actors and Experiences: Starlink and OneWeb offer bulk data transport services for data from various applications, while Kepler provides coverage to IoT devices operating beyond the reach of terrestrial networks. There are numerous constellations such as PlanetScope, Sentinel, Landsat, and ICEYE for Earth observation services. This data type has applications in agriculture, forestry and land use, disaster management, and environmental monitoring. While this type of traffic can be more delay-tolerant, the large volumes of data generated by high spatial, temporal, and spectral resolution sensors create a bottleneck at the final link to the ground station (feeder link), which remains the main limitation for these services [7].

Core Challenges: Most Earth observation constellations use RF links to communicate with ground stations. The Sentinel satellites use X and Ka bands, Landsat utilizes the X band, and PlanetScope employs S or X bands. However, the RF spectrum is becoming increasingly congested, and despite regulatory measures, the risk of interference continues to grow. Additionally, the data rates achievable with RF links are limited compared to the requirements generated by the large volumes of data involved in Earth observation [8]. On the other hand, LEO satellites have contact opportunities with each ground station that last a few minutes. If it is impossible to download the data entirely during one contact, the satellite must keep the data in its buffer and complete the download

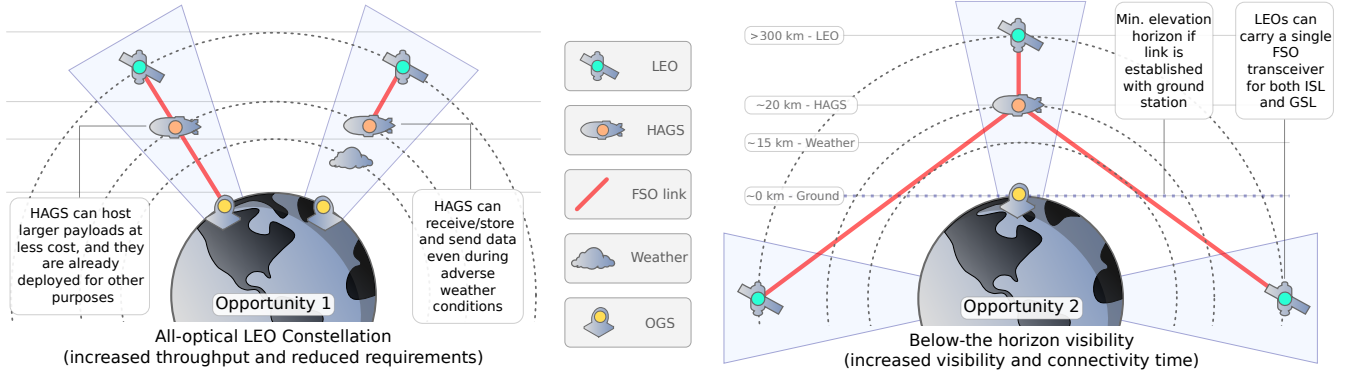


Fig. 2: Opportunities of HAPS for the bulk data transport use case: reduced weather sensitivity and increased visibility time. Figure extracted from [11] (IEEE International Conference on Communications Workshops, 2024).

during the next contact. Due to the limited buffer size on board the satellites, new data cannot be collected until the existing data is cleared. Furthermore, because RF signals are transmitted using wide beams, this type of communication is susceptible to eavesdropping, where unauthorized receivers can intercept and attempt to decrypt the signals.

Optical Feeder Links: Recently, initiatives such as the European Data Relay Satellite System (EDRS), which uses two geostationary satellites, have proposed the use of high data rate FSO links to communicate not only between satellites via optical ISLs but also between satellites and ground stations [9]. Similarly, Telesat Lightspeed plans to use optical feeder links in LEO constellations. However, one of the main challenges with FSO is their high susceptibility to adverse weather conditions, particularly cloud cover. In cloud blockages, transmission between LEO satellites and optical ground stations (OGS) becomes highly improbable. Mitigating this issue through site diversity requires deploying and maintaining multiple OGS, which makes this option less appealing. Additional challenges include the need for precise acquisition, tracking, and pointing (ATP) systems to establish and maintain links between nodes whose positions are constantly changing [10].

III. OPPORTUNITIES ENABLED BY HAPS

This section introduces a new architecture that leverages high-altitude ground stations (HAPS) in conjunction with all-optical LEO constellations. HAPS enhances bulk data transport capabilities and significantly improves direct-to-device communication performance by reducing path loss and mitigating Doppler effects. We explore the diverse opportunities this architecture offers to address the outlined challenges. A summary of all challenges and opportunities is provided in Table I.

High-Altitude Ground Stations: Considerable research has been conducted on including high-altitude platform stations (HAPS) as a critical component of 6G networks [12]. The World Economic Forum has recognized HAPS as a game-changer for global communications and one of the top 10 emerging technologies of the year across all fields [13]. According to the International Telecommunication Union (ITU),

a HAPS is a station on an object at 20 to 50 km altitude, maintaining a fixed position relative to the Earth. When HAPS is used as a base station for cellular communications, it is referred to as a HIBS, a HAPS as an international mobile telecommunications (IMT) base station [3]. In this work, we propose a new subcategory of HAPS that extends beyond the capabilities of HIBS. This novel category utilizes HAPS for RF-based access links in direct-to-device communications and backhauling IoT and Earth observation data via FSO feeder links [14]. We term this new node type a high-altitude ground station (HAPS). HAPS allows the seamless forwarding of data between LEO satellites and the ground using purely FSO links, enabling full integration with all-optical LEO constellations. At the same time, the shorter distance in direct-to-device communication enhances performance, supports localized HAPS deployments, and enables greater scalability.

A. Bulk Data Transport Over All-Optical Constellations:

Having LEO constellations that rely solely on optical transceivers for satellite-to-ground and satellite-to-satellite communications has numerous advantages. The study conducted by Airbus for the European Space Agency [15] outlines use cases, architectures, and a strategic roadmap for such constellations. Using only FSO transceivers offers significant advantages over hybrid architectures that equip LEO satellites with both RF and FSO transceivers. These include reduced size, weight, and power (SWaP) requirements, lowering the complexity and cost of launch and maintenance. Additionally, they offer enhanced security [16], a reduced risk of interference due to narrower beams, and higher data rates ranging from Gbps to even Tbps.

On the feeder link side, the main limitations of using optical links are due to cloud cover and other atmospheric disturbances. Our proposed architecture addresses these constraints by integrating HAPS positioned at an altitude of 20 km, bringing the opportunities depicted in Fig. 2. These HAPS utilize FSO transceivers to establish seamless communication links with all-optical LEO satellites and optical ground stations (OGS). In this configuration, if weather impairments disrupt the link with an OGS, data can be stored at the HAPS

Direct to Device	Challenges	Opportunities	Bulk Data Transport	Challenges	Challenges	Opportunities
Architecture	Legacy Hybrid	Proposed	Architecture	Legacy Hybrid	Legacy FSO	Proposed
SWaP of LEO Satellites	Large	Reduced	SWaP of LEO Satellites	Large	Reduced	Reduced
Path Loss	Large	Reduced	Visibility Time	Short	Short	Increased
Throughput	Low	Increased	Throughput	Low	Increased	Further Increased
Contention	High	Reduced	Weather Sensitivity	Low	High	Reduced
Interference	High	Reduced	Interference	High	Reduced	Reduced
Doppler Shift	High	Negligible	Spectrum	Limited	Unregulated	Unregulated
Device Power Consumption	High	Reduced	Security	Eavesdropping Risk	Less Eavesdropping Risk	Less Eavesdropping Risk

TABLE I: Challenges of legacy architectures and opportunities of the proposed HACS-based architecture.

and transmitted to the OGS when conditions improve. This approach offers several advantages. Since HACS are positioned above the clouds, data can be transmitted from the LEO satellite to the HACS using the time-bounded contact opportunities. From this elevated position, data can be relayed to the ground at a rate permitted by current weather conditions without considering the LEO satellite's orbital dynamics or limited contact times. A second advantage and opportunity is the increased visibility time. At an altitude of 20 km, a HACS has line of sight access to a LEO before the OGS below, thereby extending the contact window. This additional contact time, roughly 25% longer, can be utilized to transfer more data from the LEO satellite to the HACS, therefore increasing the valuable throughput.

B. Direct to Device at Closer Distances

Communicating directly with HACS at 20 km rather than with LEO satellites at 500 km reduces path loss by 28 dB. This results in a 631-fold increase in received power at the HACS, significantly enhancing achievable throughput and reducing the devices' power consumption. Moreover, since HACS are quasi-stationary, the Doppler shift between the device and the HACS becomes negligible, eliminating a significant challenge currently highlighted in 3GPP reports regarding the integration of terrestrial and non-terrestrial networks [14]. Deploying HACS over major cities also reduces the need for LEO satellites to manage contention from numerous devices, allowing them to capitalize on the aggregation capabilities of multiple HACS, which then use high-speed optical links to relay data to an all-optical LEO constellation.

IV. EVALUATION

This section quantifies the advantages of incorporating HACS into an existing LEO-OGS network, which can support both Direct-to-Device communication and Bulk Data Transport use cases. The study assumes predictable time-bounded contact opportunities between LEO satellites and HACS. However, there is uncertainty regarding the contacts between HACS and OGS due to the influence of weather conditions. Unlike previous works, we do not consider rerouting processes from HACS to the LEO satellites, as the bottleneck lies between these nodes. Instead, our solutions adopt a "prevention is better than cure" approach, prioritizing sending packets to the best available HACS rather than rerouting in case of weather blockage. This evaluation aims to build on this

foundation by achieving two objectives. The first is to evaluate the performance of the HACS-based architecture in a simple scenario involving one LEO satellite and multiple OGS. The second is to enhance these results by incorporating multiple LEO satellites connected with ISLs and weather considerations into our routing scheme.

Architecture: In this HACS-based model, each LEO satellite is equipped with an FSO terminal, which enables direct data transmission from the satellite to the HACS and to other LEOs. This facilitates continuous and seamless data sharing between LEOs at all times. We assume the predicted weather information is available when the routing solution requires it. This assumption is reasonable, as Earth-based weather data can be easily transmitted to the LEO constellation. While more precise weather data could further enhance routing performance, a reasonable level of precision is sufficient, allowing for infrequent updates.

Cloud Coverage Model: We employ an exponential distribution to model the weather events, which captures cloud coverage's random and memory-less nature. This distribution is characterized by the mean time to cloud coverage (TCC) and the mean time to clear sky (TCS).

Metrics: The performance is evaluated using three key metrics: *a)* Delivery Ratio: measuring the percentage of files that are successfully delivered to the final destination, *b)* Delivery Delay: calculated as the difference between file generation time and arrival time, and *c)* Buffer Occupation: characterized by mean and maximum values, computed as an average at each HACS.

A. Single LEO - Multiple OGS

When focusing on a single LEO, the aim is to determine how many HACS-OGS pairs are necessary to achieve the same performance as a basic LEO-OGS architecture, considering an increasing and larger number of OGS. Consequently, the delivery ratio and delivery delay are calculated for the LEO-OGS architecture with one, two, five, and ten OGS, as well as for one to five HACS-OGS pairs. In addition, the weather conditions are varied with the TCC and TCS parameters. The TCC ranges between 0.1 and 40 hours, and the TCS ranges from 5 to 25 hours. To achieve the intended equivalence, it is first necessary to identify the points of intersection between all of the curves representing the delivery ratio and the delivery delay across all simulation iterations.

The results of the generalized equivalence are illustrated in Fig. 3. The illustration demonstrates that, in most cases, a

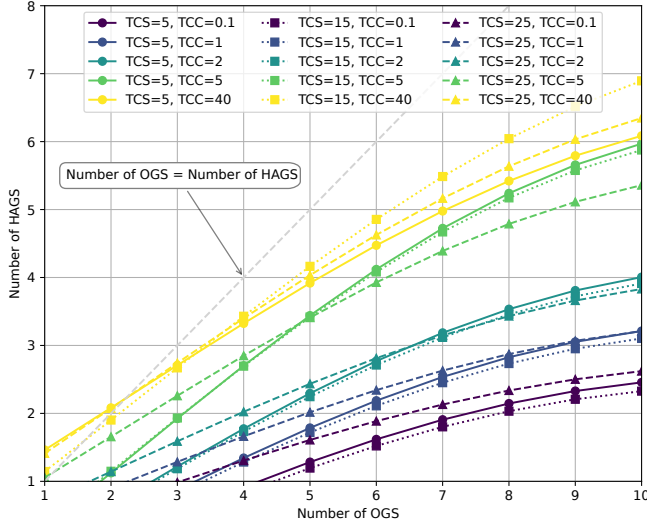


Fig. 3: Equivalency analysis between OGS and HAGS. Figure extracted from [11] (IEEE International Conference on Communications Workshops, 2024).

smaller number of HAGS can achieve the same performance as a larger number of OGS. For example, two HAGS were capable of delivering performance on par with eight OGS for TCS ranging between 5 and 15 hours and a TCC of 0.1. Notably, as weather conditions improve (characterized by longer TCC and shorter TCS), the advantage of using HAGS diminishes. However, they remain a more efficient alternative than a larger array of OGS, particularly for connecting all-optical LEO satellites.

B. Multiple LEO - Multiple OGS

When multiple LEOs are considered, the routing scheme plays an important role in how data is forwarded to the destination. In this work, we compare three routing scheme variants.

1) *Greedy Routing*: The *greedy* routing approach selects the best route based on arrival time for each packet, regardless of the weather. Therefore, it tries to get the maximum from each contact by only using global topological information.

2) *Equal Routing*: Our initial contribution here is an equal distribution routing scheme. This approach ensures that each HAGS receives an equal share of packets, as we observed that the greedy routing solution often directs many packets to a single HAGS. While this routing solution does not explicitly consider weather conditions, it assumes uniform cloud coverage across each HAGS-OGS pair. Notably, the implementation is straightforward, requiring only minor modifications compared to the greedy solution. We opted for a round-robin distribution, routing one packet to each HAGS individually rather than batch-processing them.

3) *Weighted Routing*: To properly account for the weather, we developed a solution that calculates the fraction of clear sky in a HAGS by dividing the mean time to cloud cover (equal to the mean time of clear sky) by the sum of the mean time to cloud cover plus the mean time to clear sky. Then,

the percentage of packets that need to be sent to a HAGS is calculated by dividing the proportion of clear sky at that HAGS by the total sum of clear sky in all the HAGS.

Results: The results of this evaluation are presented separately for each metric in Figs. 4 and 5. The x-axis of the graph represents the ratio between the more and less cloudy areas. In the examples shown, the cloud cover ratio reaches 40, meaning the cloudier area experiences 40 times more cloud cover than the less cloudy area. In the following, we discuss the aspects of the delivery ratio, delivery delay, and buffer occupation.

a) *Delivery Ratio*: Fig. 4 illustrates the evolution of delivery ratios for varying ratios of cloud coverage. We observe that both the equal and weighted routing solutions outperform the greedy routing. This is because packets keep getting stored in HAGS after the simulation ends, as they are sent opportunistically, implying that some HAGS received significantly more packets than their clear sky percentage. For relatively low values of the cloud cover ratio, the equal and weighted solutions are essentially indistinguishable. This is expected, as the cloud coverage is nearly identical, leading to similar routing decisions for both approaches. For larger values of

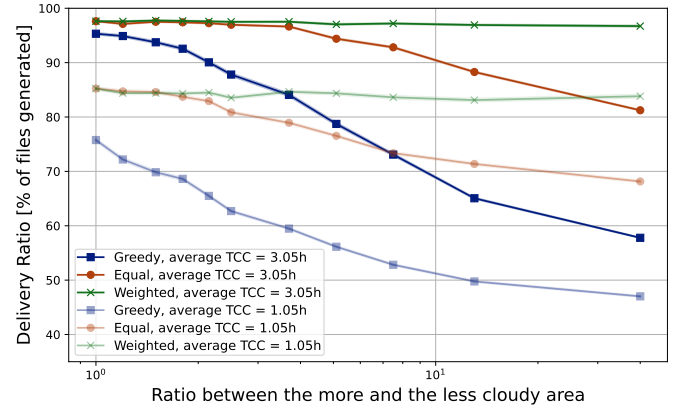


Fig. 4: Delivery ratio for each routing solution. Figure extracted from a paper submitted to the IEEE Middle East Conference on Communications and Networking, 2024.

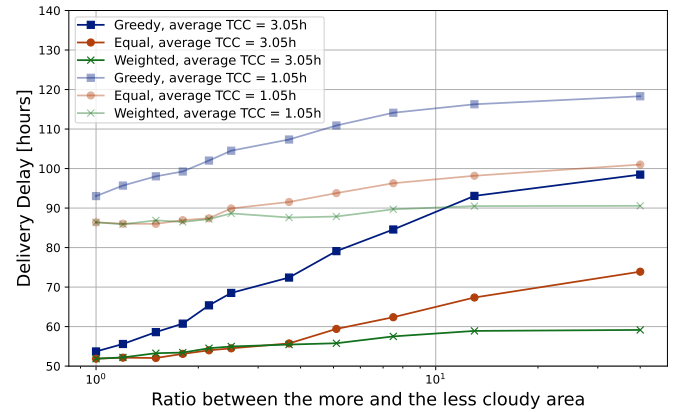


Fig. 5: Delivery delay for each routing solution. Figure extracted from a paper submitted to the IEEE Middle East Conference on Communications and Networking, 2024.

the ratio of cloud cover, the equal routing solution continues to distribute packets evenly among every HAGS, leading to a performance loss due to the same reason as greedy routing. It's worth noting that the weighted distribution's performance remains consistent, demonstrating that the solution works effectively by utilizing each HAGS at its full potential. The 100% delivery ratio was not achieved only because the average availability of all OGS was insufficient to allow all packets to reach their destinations.

b) Delivery Delay: The delay comparison requires caution when examining Fig. 5, as the delay value is only recorded when the packet reaches the destination node. To compare the scenarios, we have artificially added all the remaining packets to the simulation with a delay of the total duration of the simulation. This approach is not entirely accurate, as the delay in reaching the final node would likely be more significant. However, it provides a reasonable basis for comparing the different routing solutions. As expected, the greedy routing solution takes much longer on average to deliver the packets. For a small ratio, equal and weighted are close, and for a larger ratio, weighted becomes increasingly appealing by keeping the delay constant as the ratio increases, again showing the use of each HAGS to its full potential. This shows that the weighted routing solution has a better delivery ratio with less delay than the other schemes.

c) Buffer Occupation: The findings demonstrate that the weighted routing solution achieves up to a 15% reduction in buffer occupancy compared to the equal routing approach, and up to 25% compared to the greedy solution. By factoring in the average weather conditions for each HAGS-GS pair, the weighted routing dynamically adjusts packet transmission rates to align with each HAGS's actual download capacity.

V. CONCLUSION

This article presented a novel NTN architecture integrating high-altitude ground stations (HAGS) with all-optical LEO constellations. This architecture addresses the critical challenges of traditional hybrid and free-space optical systems, providing a more efficient and scalable solution for both direct-to-device and bulk data transport use cases. The proposed approach reduces the LEO satellite platform's size, weight, power consumption, and cost. HAGS located at 20 km altitude extend the contact window by approximately 25%, increasing throughput for data transfer. Our analysis shows that two HAGS can match the performance of eight optical ground stations (OGS), demonstrating the efficiency of HAGS in managing data traffic. Additionally, our weighted routing strategy results in a reduction in buffer occupation of up to 25%, ensuring better load distribution and improved system efficiency. These advancements make connecting remote and underserved areas more attainable and offer a cost-effective and sustainable solution for the future of satellite constellations, potentially boosting network profitability and driving widespread adoption.

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